

Lord of the recommendation models, Two Tower for All(egro)



Aleksandra Osowska-Kurczab
Machine Learning Manager



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Senior Data Engineer

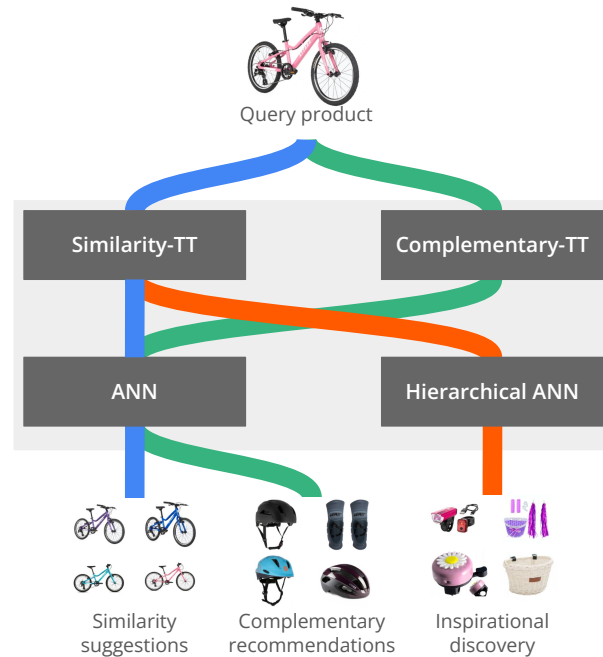


Paweł Młyniec
Junior Research Engineer

Allegro
Tech Talks #45
AI edition
-14.01.2026-

Agenda

- 1 Recommendations 101
- 2 Information retrieval
- 3 Allegro use cases
 - a. Similarity search
 - b. Complementary recommendations
 - c. Inspirations
- 4 Deployment scenario



Osowska et al.
*Suggest, Complement, Inspire: Story of Two
Tower Recommendations at Allegro.com*



What is a recommender system?

Recommender – algorithm-driven system that helps suggest relevant items to users, based on their behavior, preferences, or what similar users have liked.





Film: Forrest Gump

Avg score: 8.8★

Year: 1994

Duration: 2h 22m

My score: 8★

More like this



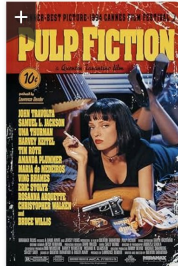
★ 8.8 ☆

Fight Club



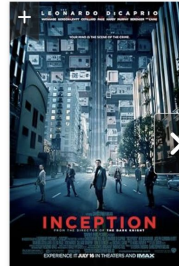
★ 9.3 ☆

The Shawshank Redemption



★ 8.9 ☆

Pulp Fiction



★ 8.8 ☆

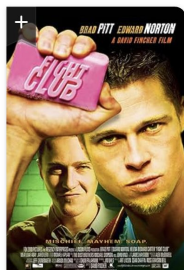
Inception

IMDb



Film: Forrest Gump
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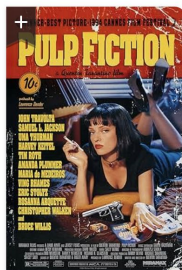
More like this



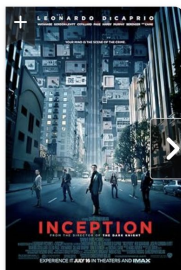
★ 8.8 ☆
Fight Club



★ 9.3 ☆
The Shawshank Redemption



★ 8.9 ☆
Pulp Fiction

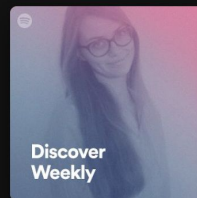


★ 8.8 ☆
Inception



User: Aleksandra Osowska-Kurczab
Yearly playtime: 3 000h
Favourite artist: Disturbed
Top genre: Metal

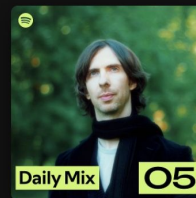
Made For Aleksandra Osowska-Kurczab



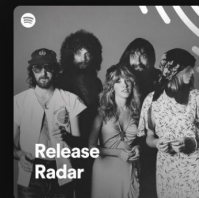
Your weekly mixtape of fresh music. Enjoy new...



Metallica, Disturbed, Rammstein and more



Oskar Schuster, Fabrizio Paterlini, Max Richter...



Catch all the latest music from artists you follow,...



Lego Petite Sunny Bouquet



4,96 ★★★★★ 87 ocen i 16 recenzji



SUPERCENA

95,90 zł SMART!



Add to cart



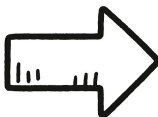
Lego Petite Sunny Bouquet

4,96 ★★★★★ 87 ocen i 16 recenzji



SUPERCENA

95,90 zł SMART!



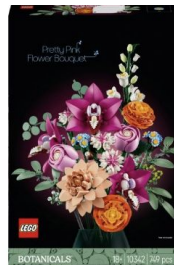
Item added to the cart



95,90 zł SMART!

Lego Petite Sunny Bouquet

Order in one package



165,90 zł SMART!

Lego ICONS Pretty
Pink Flower Bouquet

14,99 zł SMART!

Lego Botanicals Field
Flowers

80,99 zł SMART!

Lego Creator 3 in 1
Pink Flamingo

80,99 zł SMART!

Lego ICONS Min
Orchid

Add to cart

And **100+** other scenarios!

Recommendations at Allegro

Findability

Comparison

SMART 

Basket
Building

Inspiration

allegro



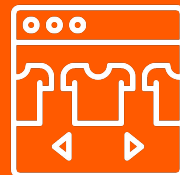
60%

Allegro users



250B+

yearly views of
reco carousels

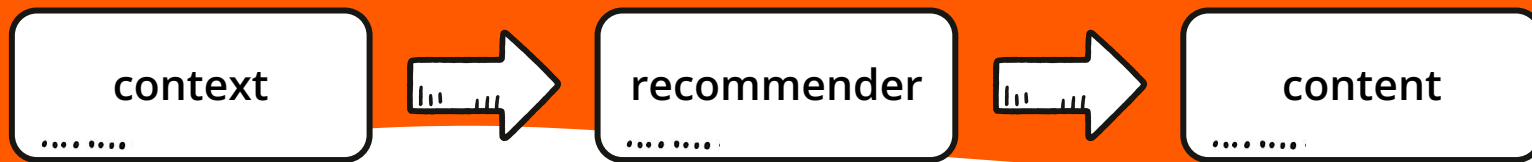


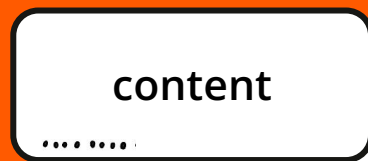
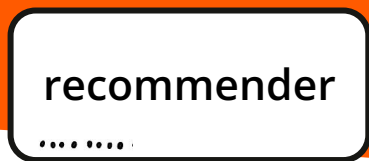
100+

recommendation
scenarios

Recommender systems under the hood







User profile



Other clients



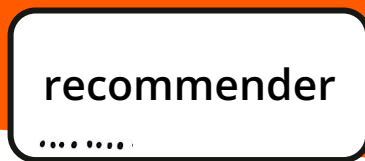
Catalogue



Current actions



Past actions



User profile



Other clients



Catalogue

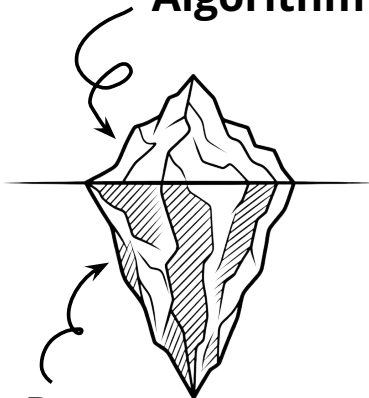


Current actions

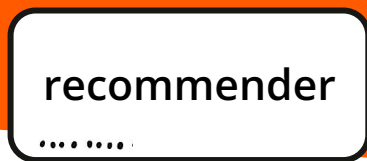


Past actions

Algorithm



Data
Business logic
Analytics
Serving
Monitoring



User profile



Other clients



Catalogue

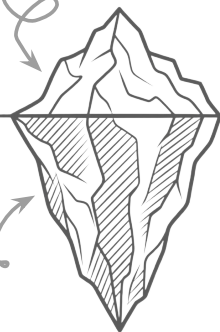


Current actions



Past actions

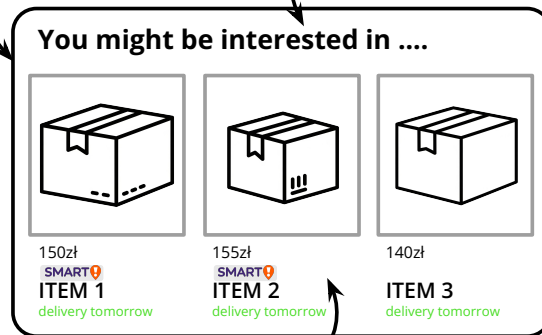
Algorithm



Data
Business logic
Analytics
Serving
Monitoring

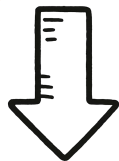
Placement

Slogan

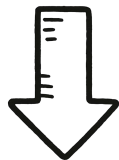


UX design

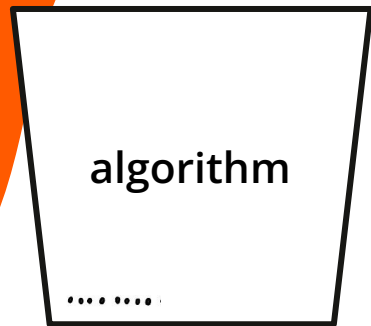
context
... ..



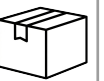
recommender
... ..



content
... ..

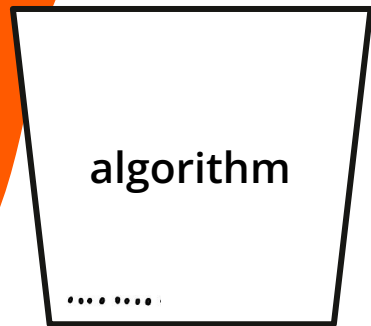


You might be interested in

		
150zł SMART ITEM 1	155zł SMART ITEM 2	140zł ITEM 3

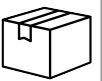


millions



dozens

You might be interested in

		
150zł SMART ITEM 1	155zł SMART ITEM 2	140zł ITEM 3



millions



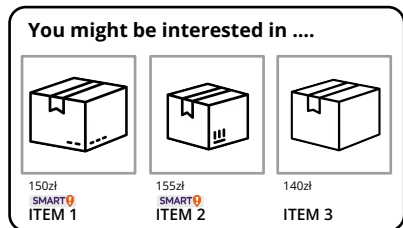
lightweight filtration of the catalogue

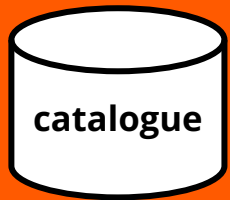


precise sorting of the candidates

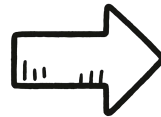
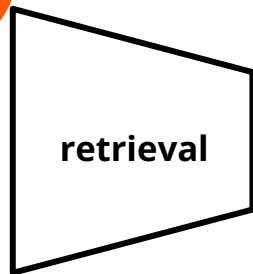


dozens

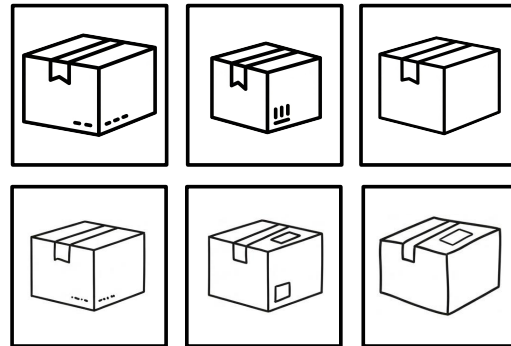




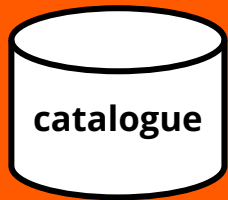
millions



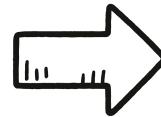
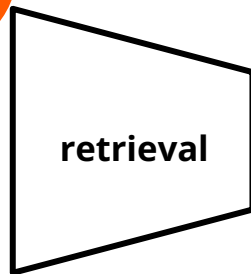
hundreds



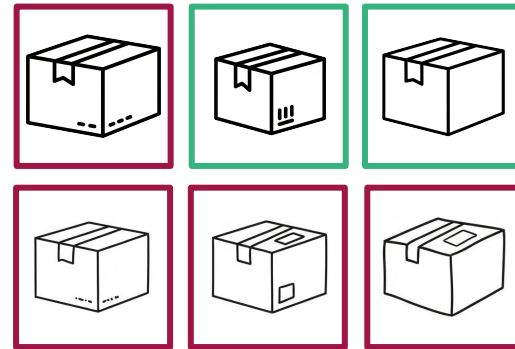
Performance
measured by
recall@k



millions



hundreds



Performance
measured by
recall@k

$$\text{recall@k} = \frac{\begin{array}{l} \# \text{ of relevant} \\ \text{recommendations} \end{array}}{\begin{array}{l} \# \text{ of relevant offers} \end{array}} = \frac{2}{6}$$

Collaborative filtering

suggests items based on what **similar people** have liked in the past

Content-based filtering

recommends things that are **similar in style or type** to what you have already enjoyed

Generative recommendations

unifies reco tasks into a single paradigm of predicting **next user actions** (sequence)

Heuristic-based recommendations

uses **simple rules** to connect items with a user e.g. bestsellers or newest arrivals

Hybrid filtering

combines multiple approaches to retrieval e.g. collaborative + content-based filtration

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Hybrid filtering

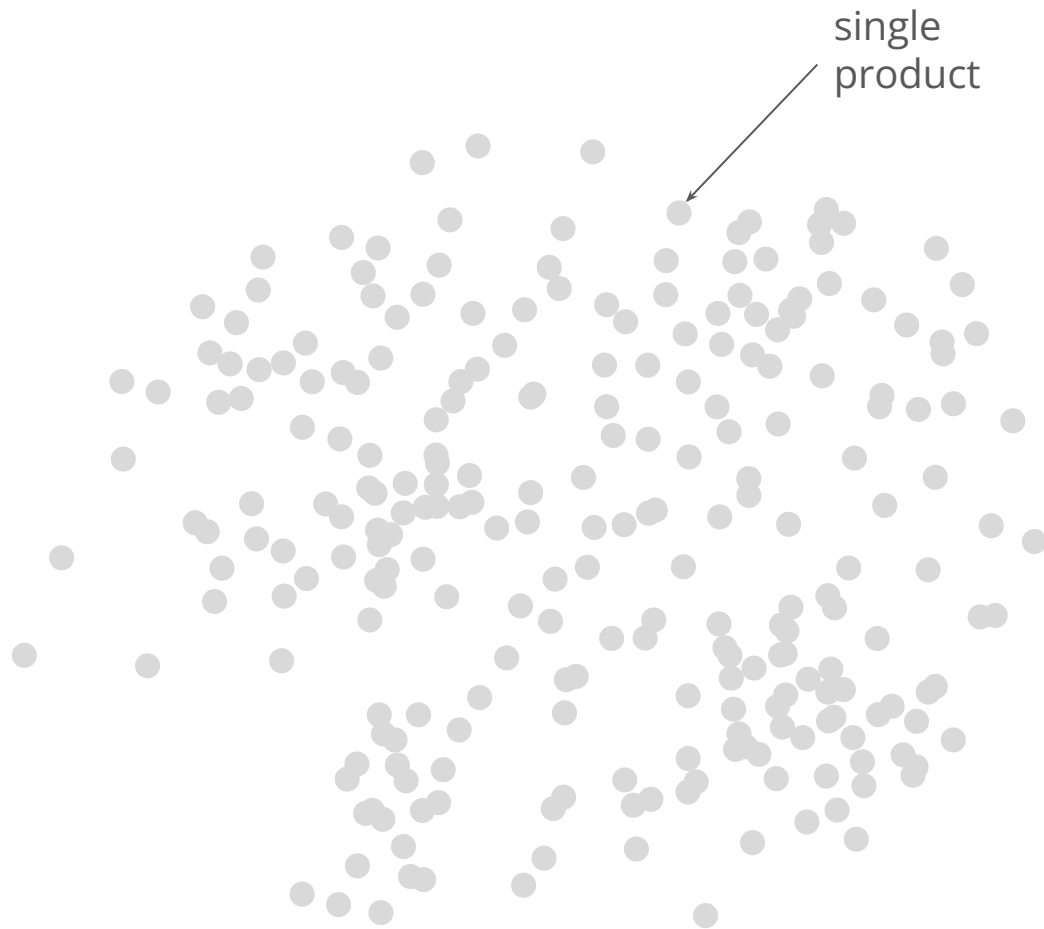
combines multiple approaches to retrieval e.g. collaborative + content-based filtration

Recipe:



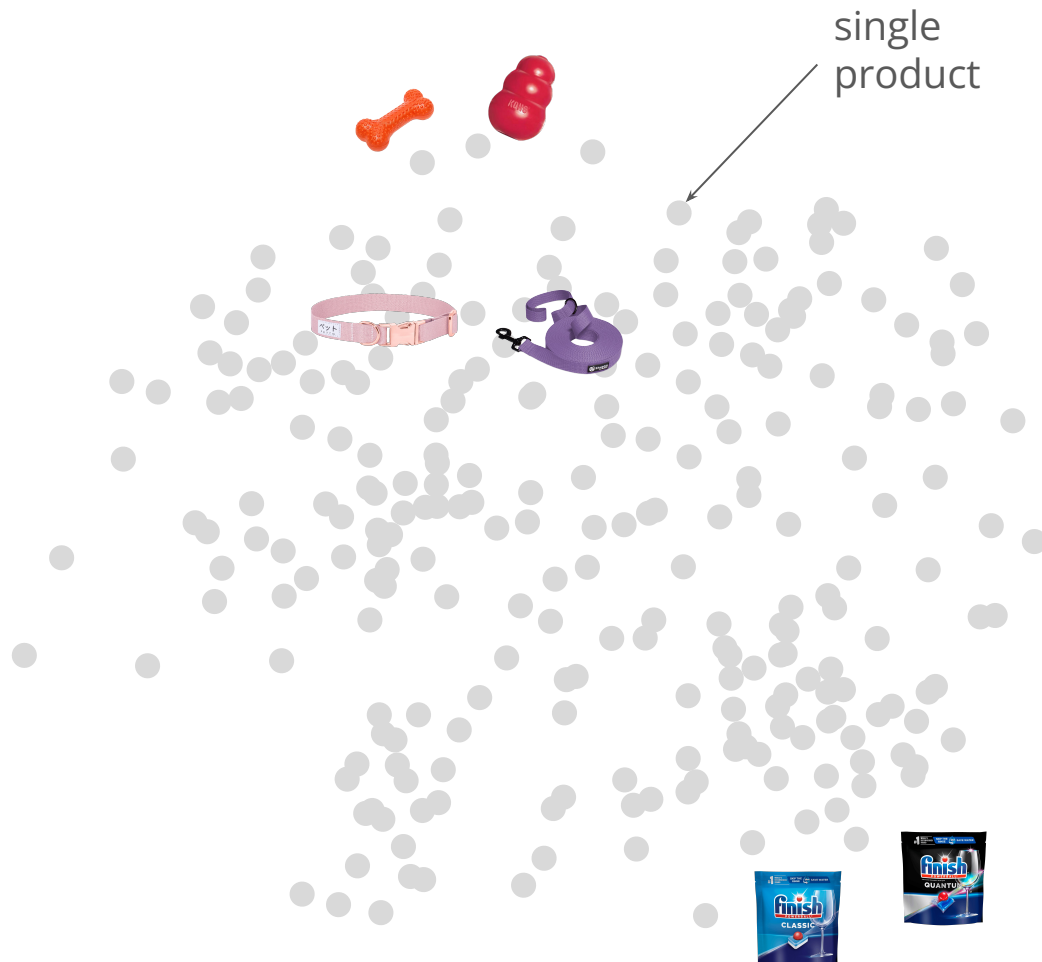
Recipe:

1. Build embedding space



Recipe:

1. Build embedding space

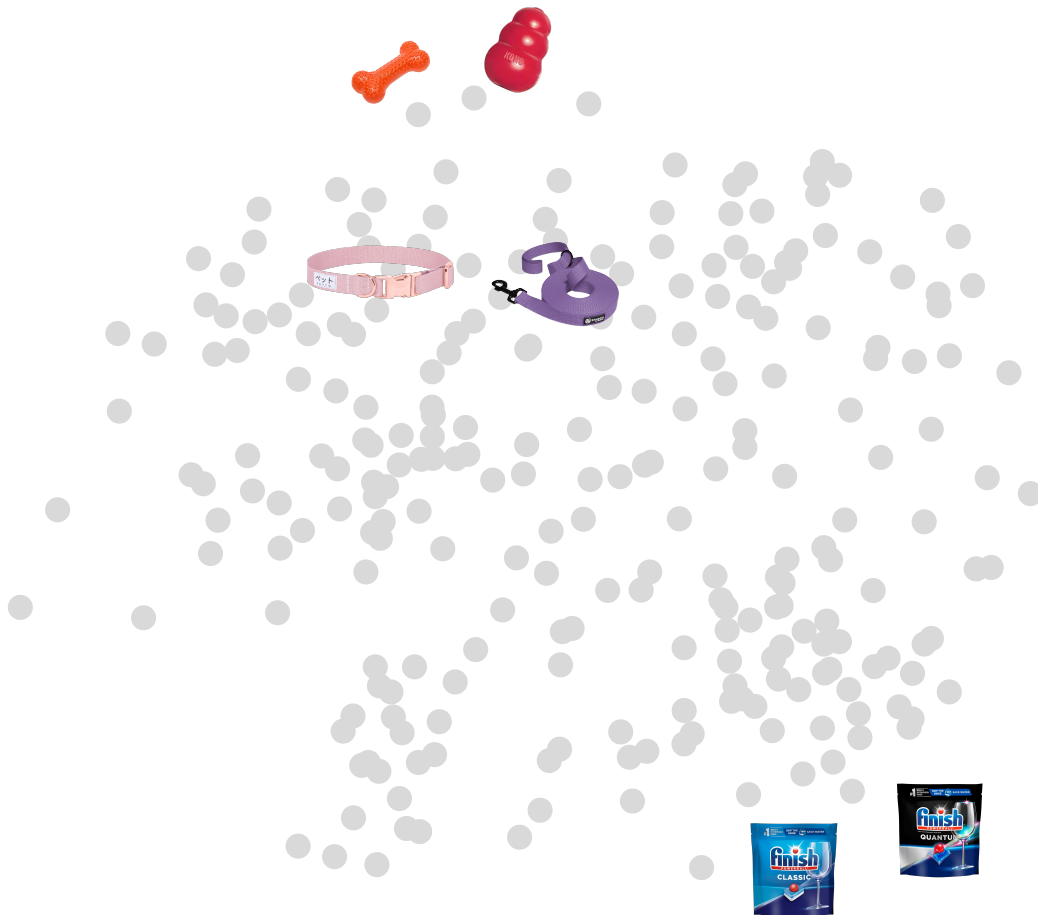


Recipe:

1. Build embedding space
2. Take the context
of your request



context



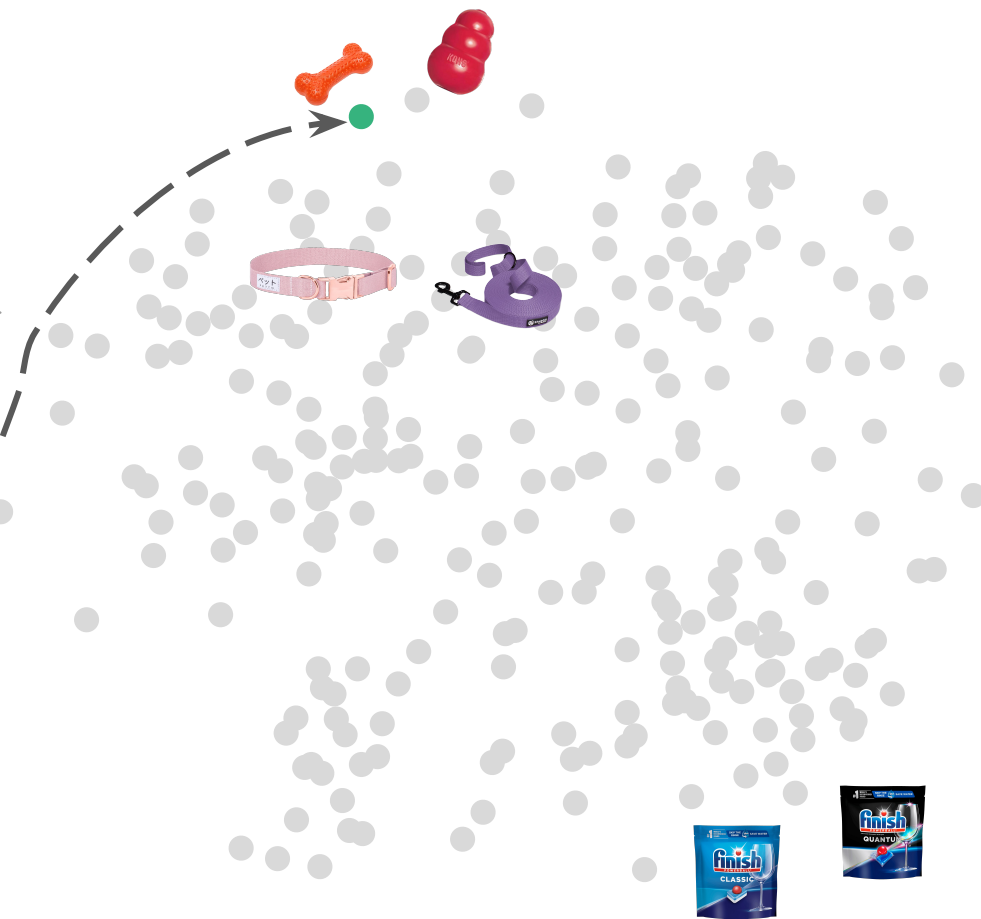
Recipe:

1. Build embedding space
2. Take the context of your request
3. Embed it via ML model to a representation



context

encoder



Recipe:

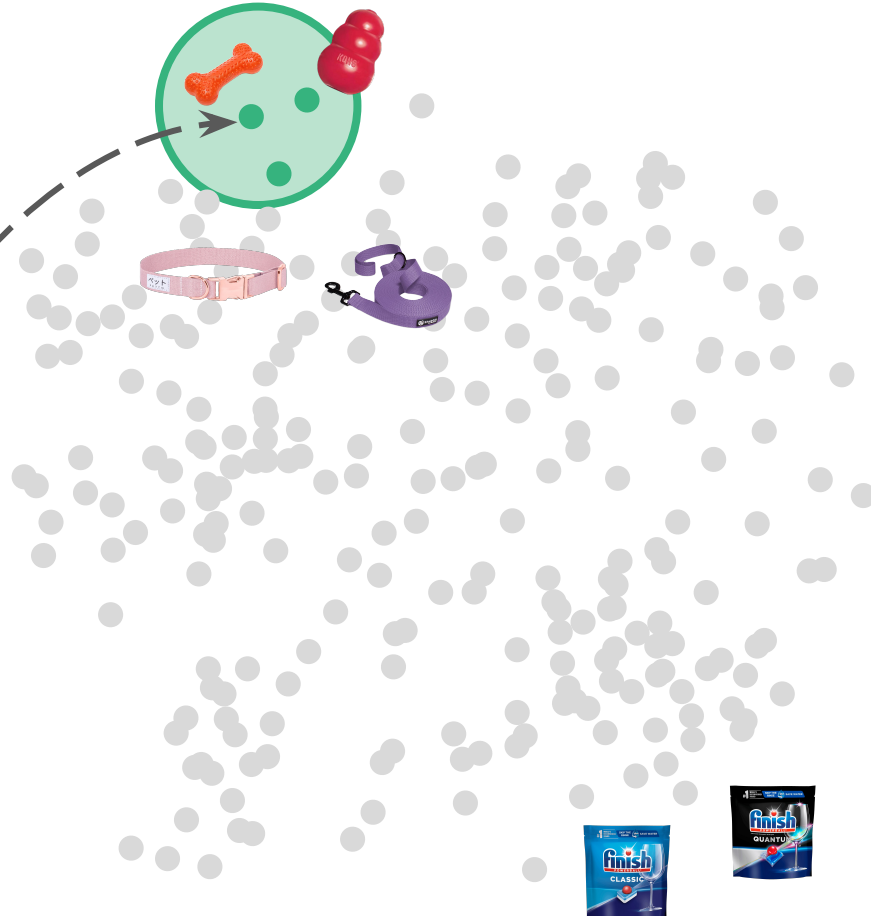
1. Build embedding space
2. Take the context of your request
3. Embed it via ML model to a representation
4. Return matching responses via ANN search



context

encoder

ANN index



A large, white, irregular blob shape is centered on a solid blue background. The blob has a soft, organic feel with rounded edges and a few small indentations. Inside the blob, the text "What is similarity search?" is written in a blue, sans-serif font.

What is similarity search?

What is similarity search?



Query product

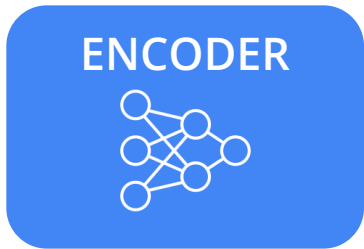


Similarity suggestions

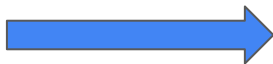
What is similarity search?



Query product



[0.4, -0.7, 0.5, 0.8]
embeddings



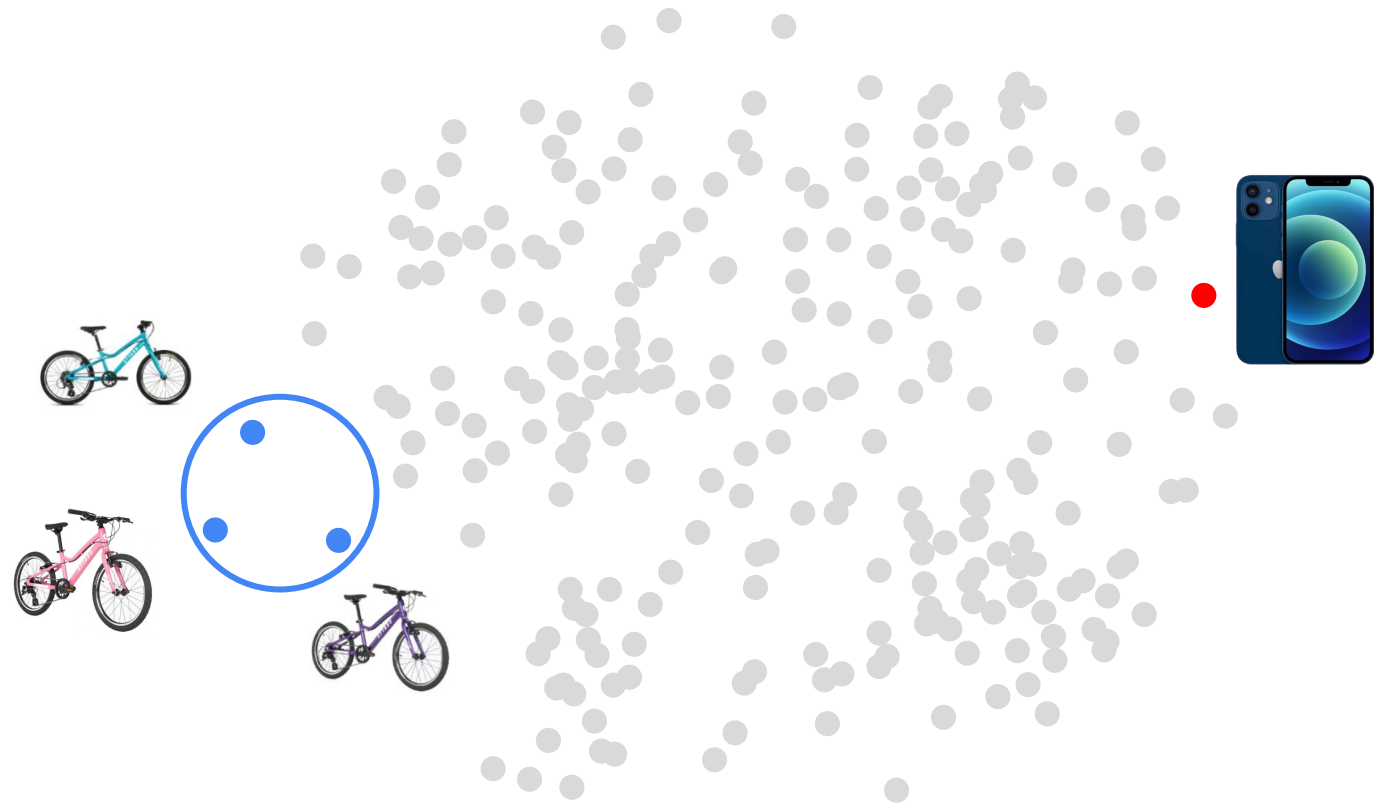
Similarity suggestions



pre-computed vector
representations



What is similarity search?



Training a model for
similarity search

query input

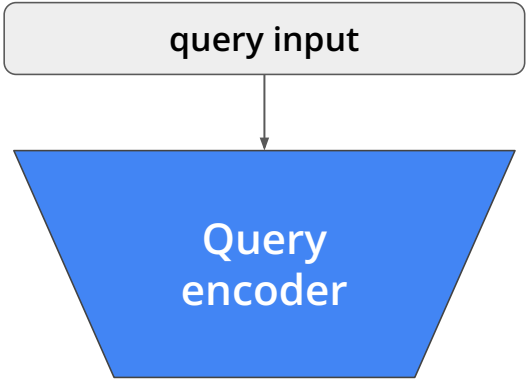
Training a model for similarity search

Our input data, product information such
as title, price, category path

query input

Training a model for similarity search

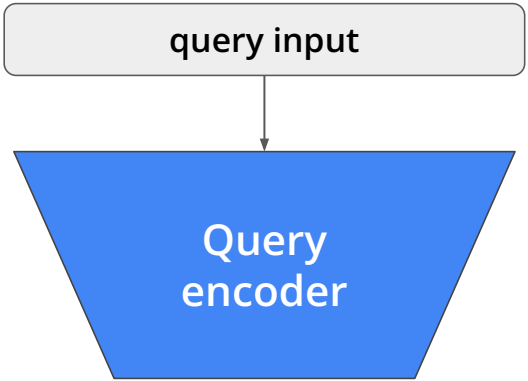
Our input data, product information such as title, price, category path



Training a model for similarity search

Our input data, product information such as title, price, category path

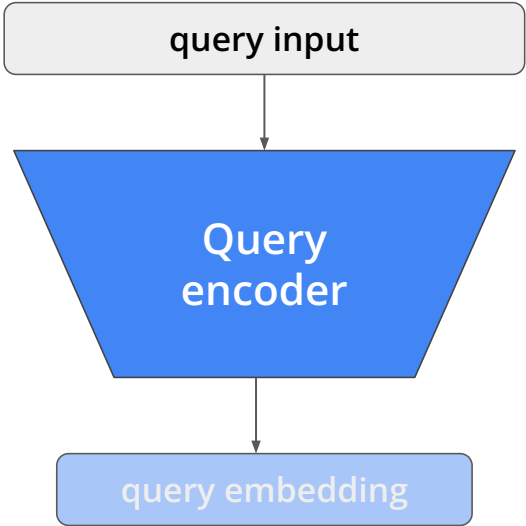
Our neural network which produces...



Training a model for similarity search

Our input data, product information such as title, price, category path

Our neural network which produces...

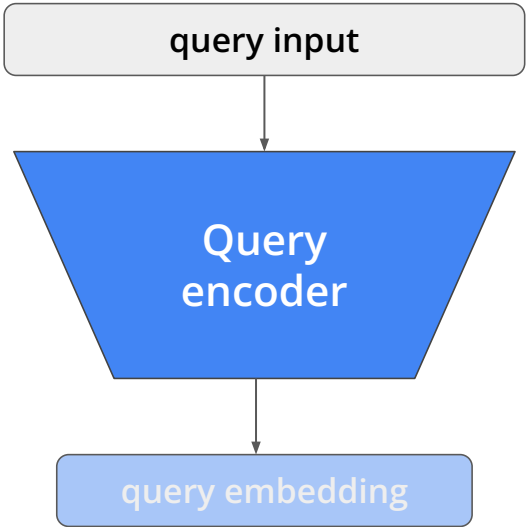


Training a model for similarity search

Our input data, product information such as title, price, category path

Our neural network which produces...

Vector representation of the input data

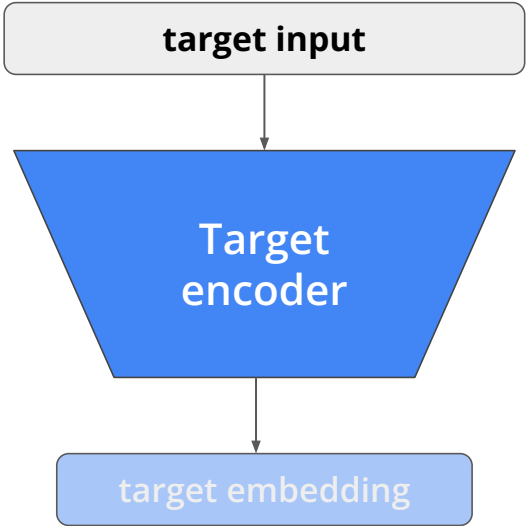


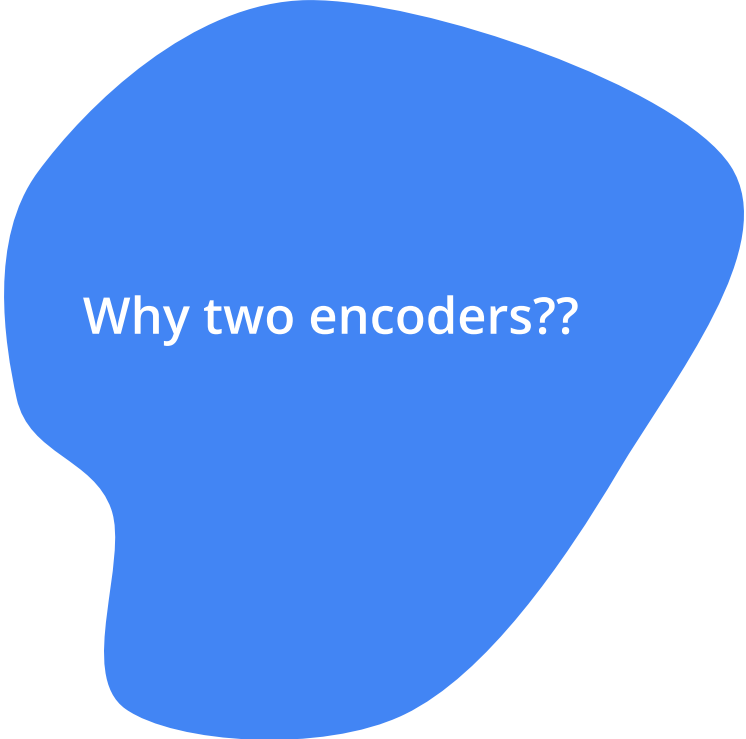
Training a model for similarity search

Our targets are also products on Allegro website

This encoder can be bigger - contain more weights and layers

Target embeddings are once again vector representations of the products

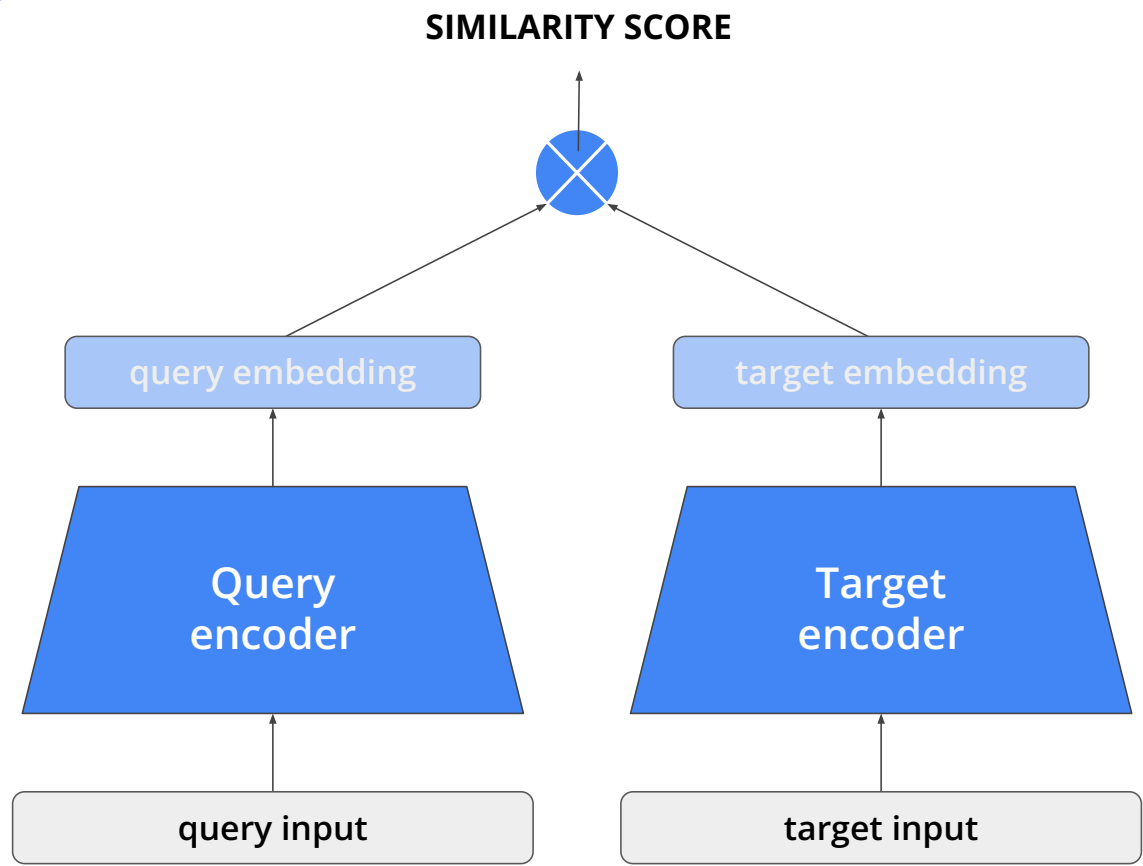


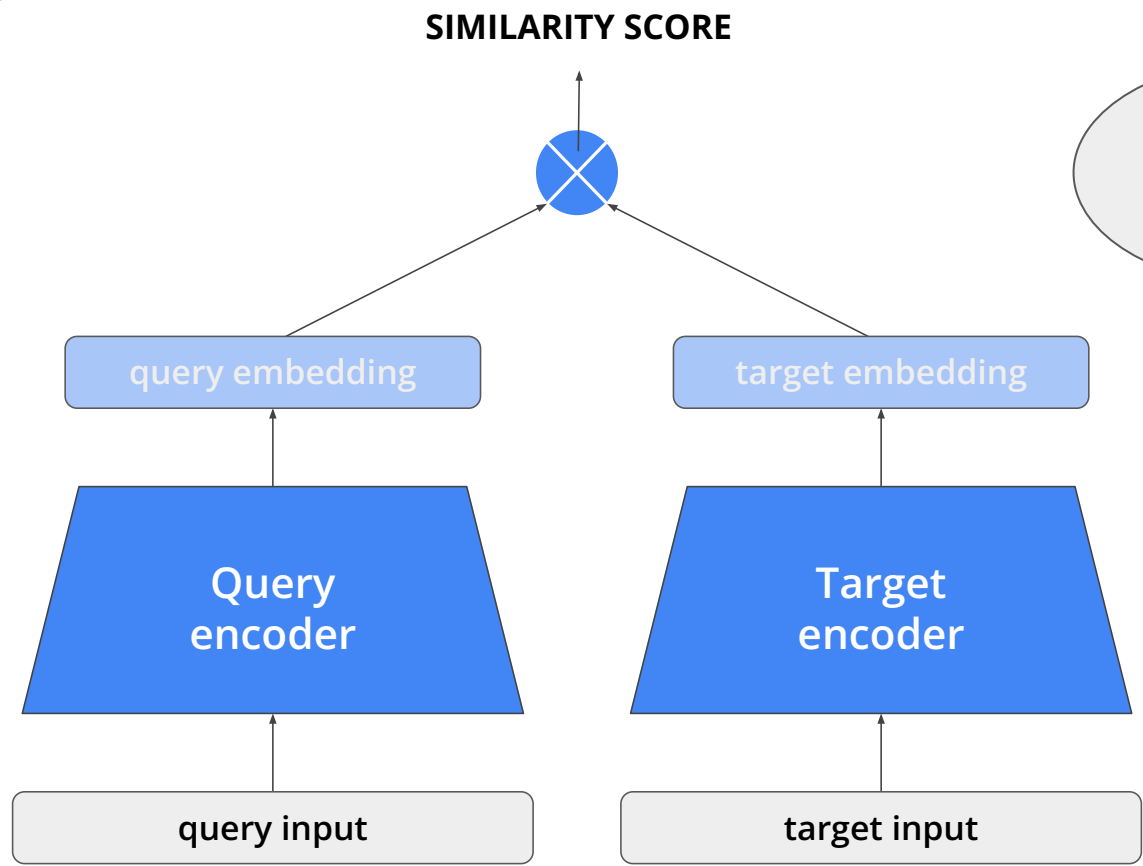
A solid blue, irregular blob-like shape with a soft, organic feel, centered on a white background. It has several rounded protrusions and indentations, resembling a splash or a cloud.

Why two encoders??

name a more iconic duo.. I'll wait.



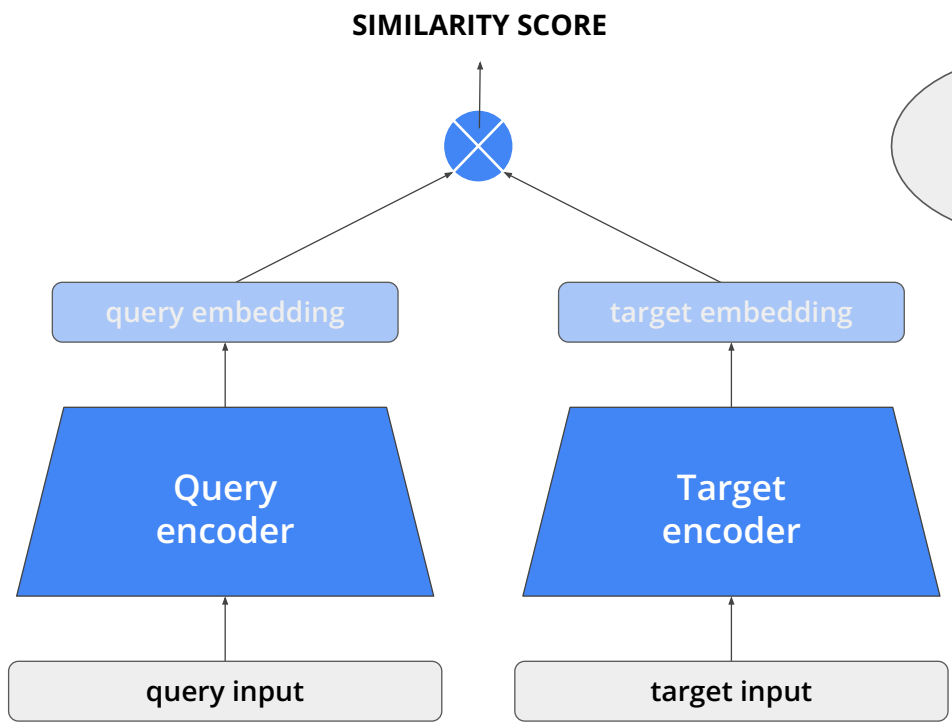




This architecture is called "two-towers"

Training a model for similarity search

Training data are co-viewed pairs of items in the same session

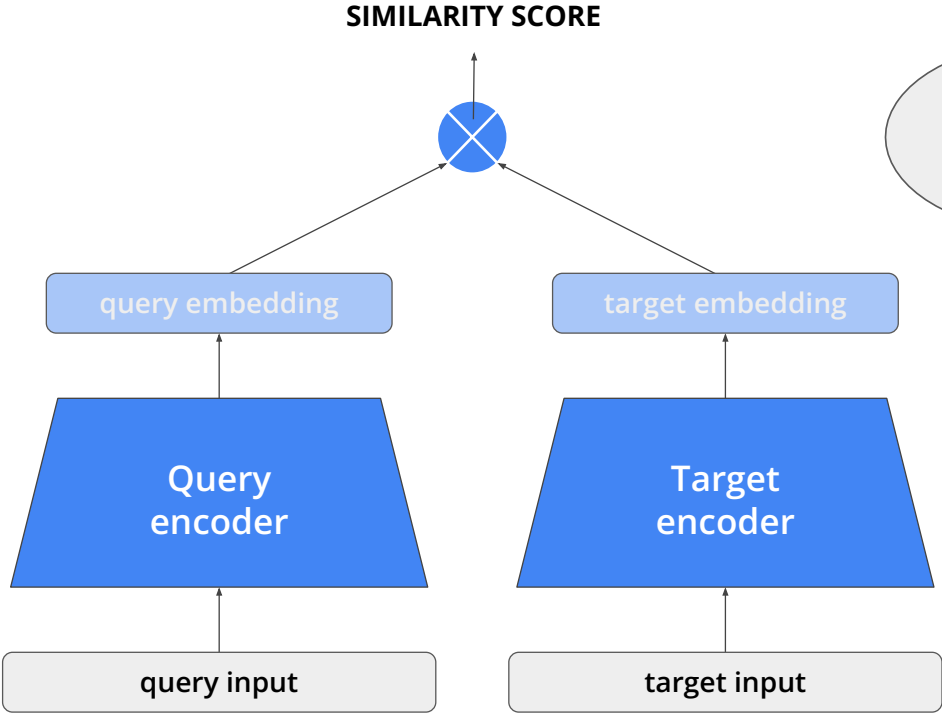


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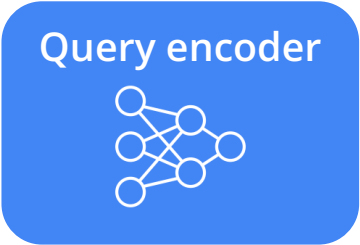
Training a model for similarity search

Training data are co-viewed pairs of items in the same session

Our goal is to maximise the similarity score between those pairs



This architecture is called "two-towers"



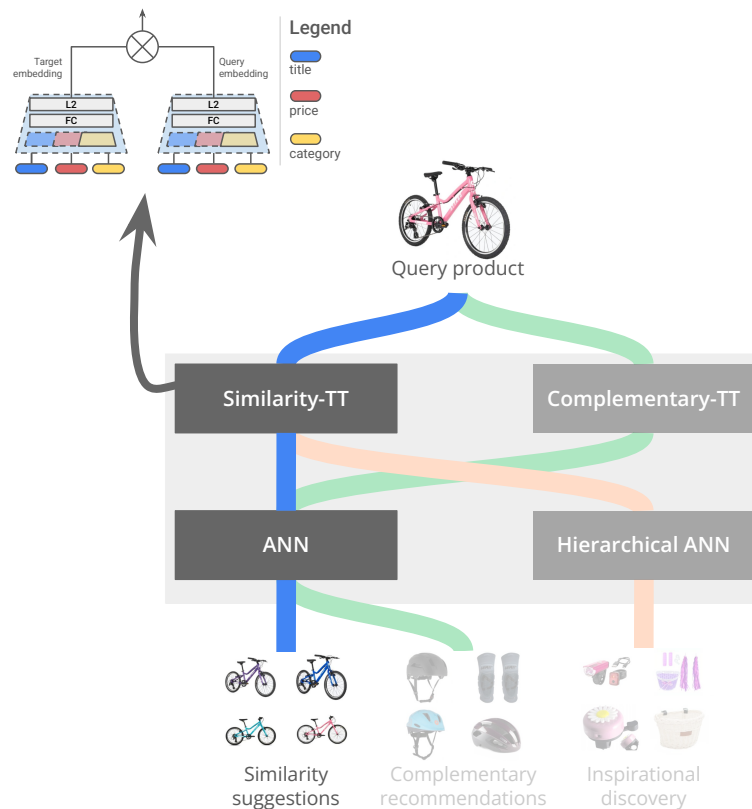
Used for encoding data
during online inference



Used for creating ANN
index

Similarity Search - recap

- What is similarity search?
- Training a model for similarity search





What are complementary
recommendations?

What are complementary recommendations?

...items that can
“complement” certain
purchase



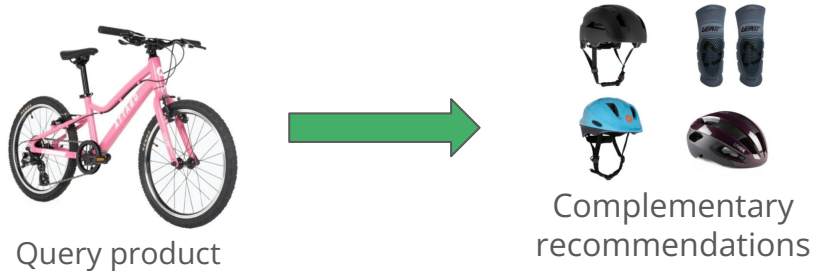
Query product



Complementary
recommendations

What are complementary recommendations?

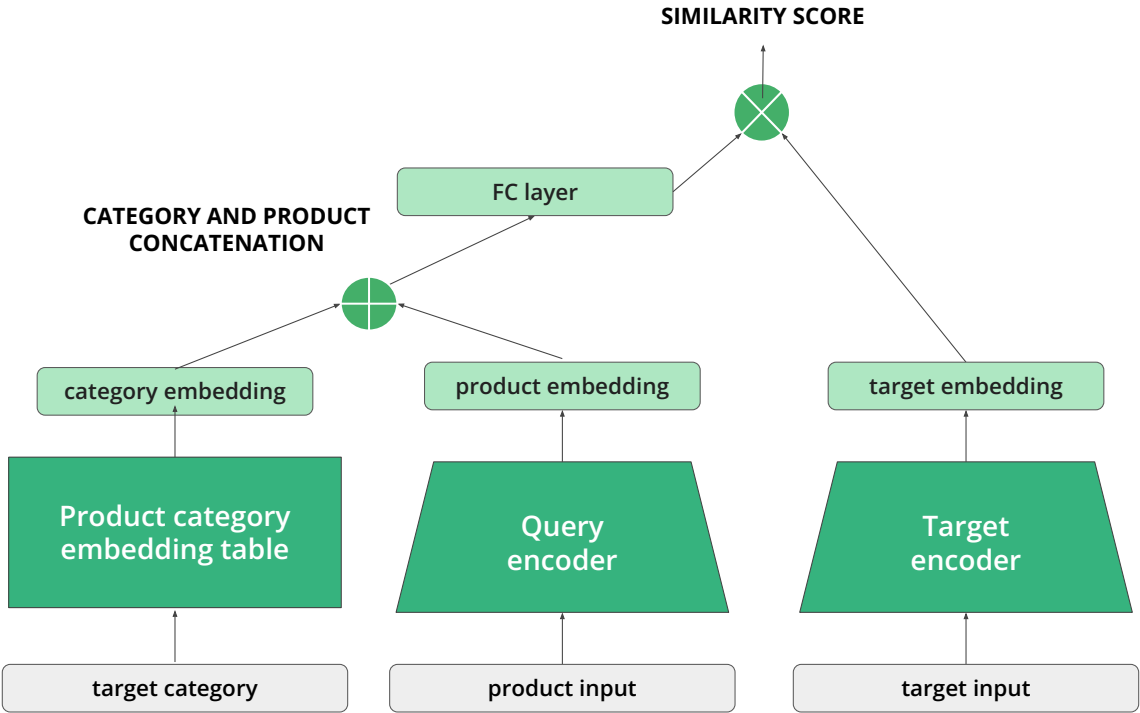
...items that can
“complement” certain
purchase



Essentially similarity search but with different targets

Training two towers model for complementary recommendations

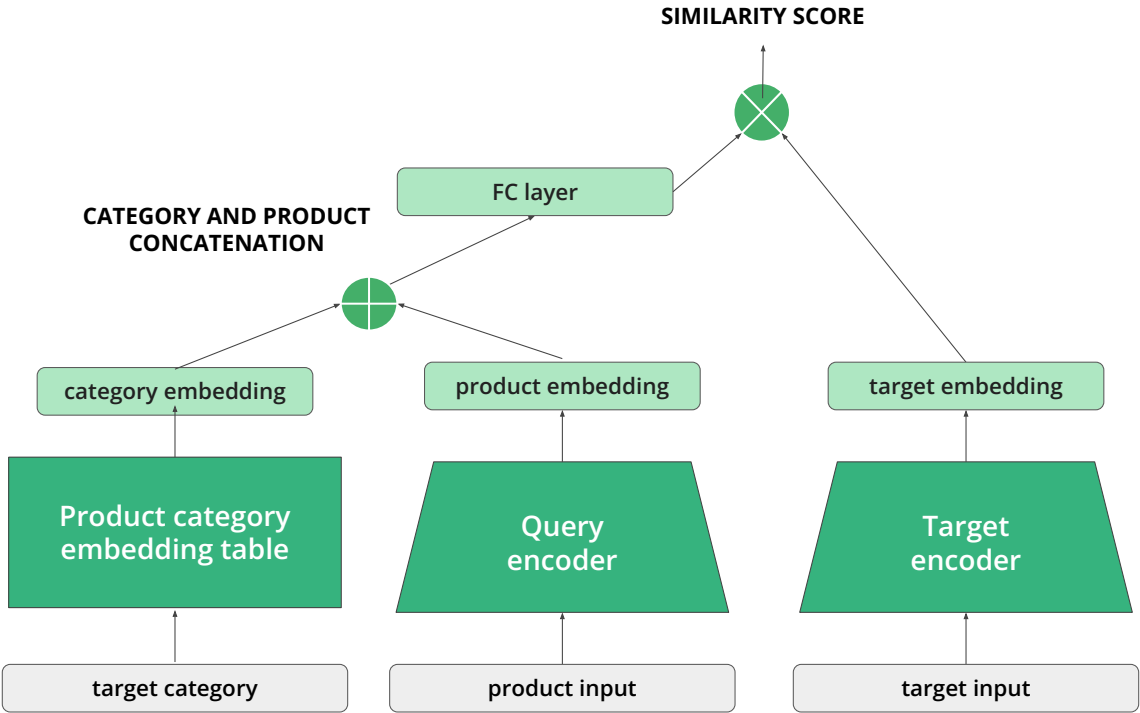
Same “two-tower” architecture



Training two towers model for complementary recommendations

Same “two-tower” architecture

This time query is a concatenation of product embedding and complementary category embedding

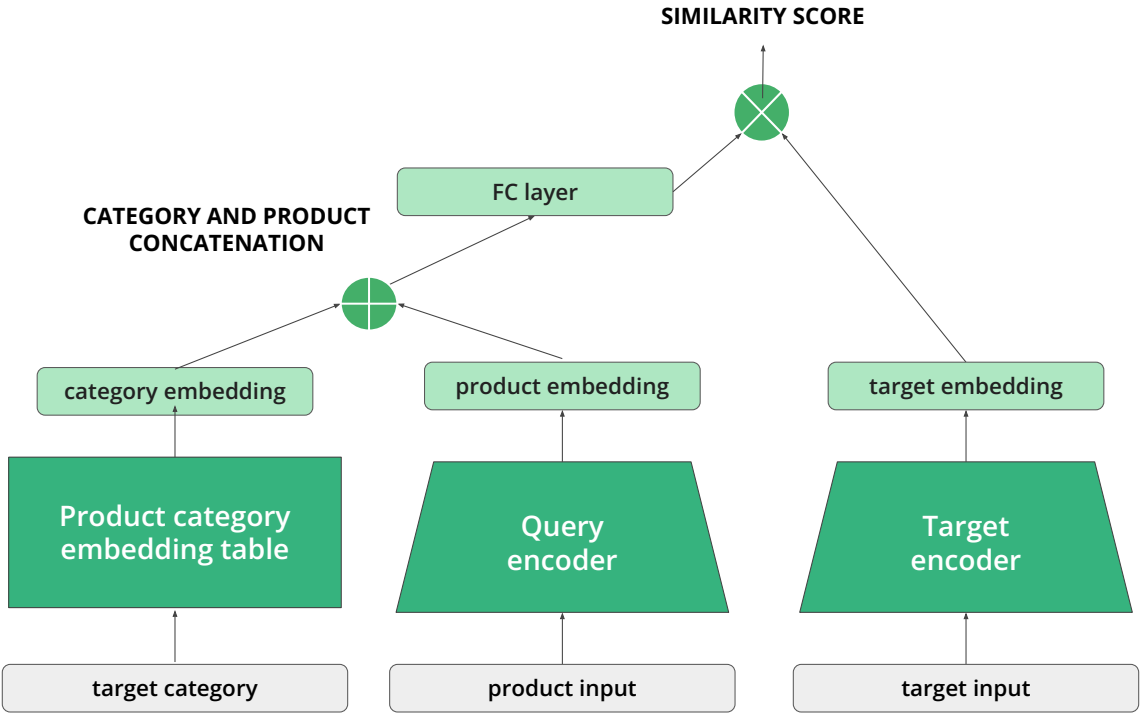


Training two towers model for complementary recommendations

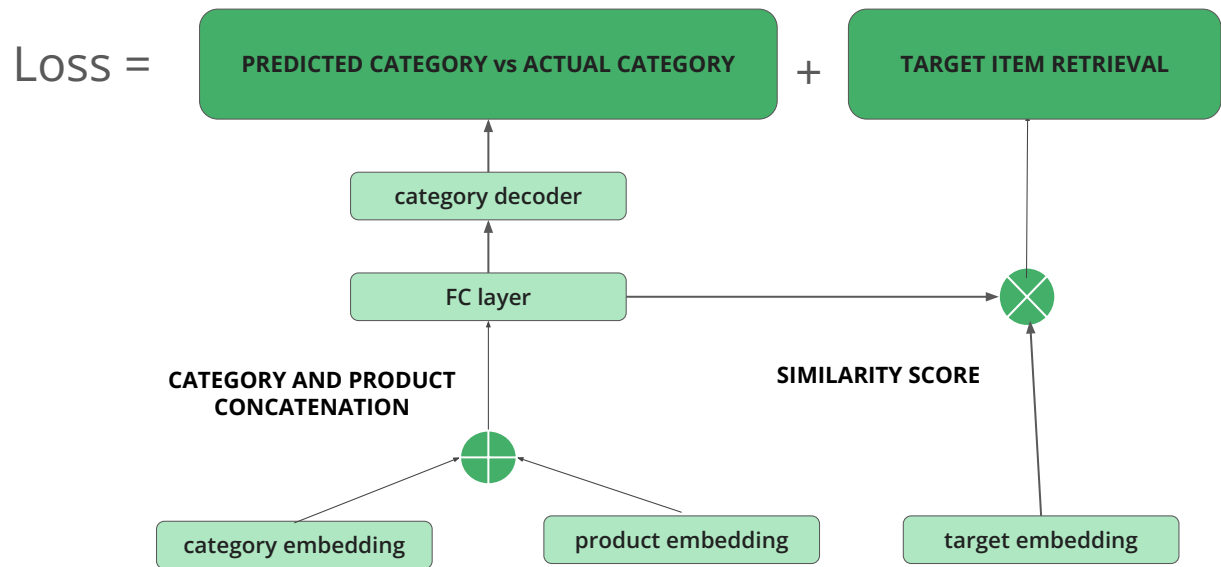
Same “two-tower” architecture

This time query is a concatenation of product embedding and complementary category embedding

Loss function is enhanced with target category reconstruction error which helps the model to learn the correct categories to recommend



Training two towers model for complementary recommendations



Training two towers model for complementary recommendations

During training we know the complementary categories from target products

Training two towers model for complementary recommendations

During training we know the complementary categories from target products

But for inference we don't have targets, as we will be predicting them, so we don't know target categories

Training two towers model for complementary recommendations

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We can use predefined complementary categories mapping

Training two towers model for complementary recommendations

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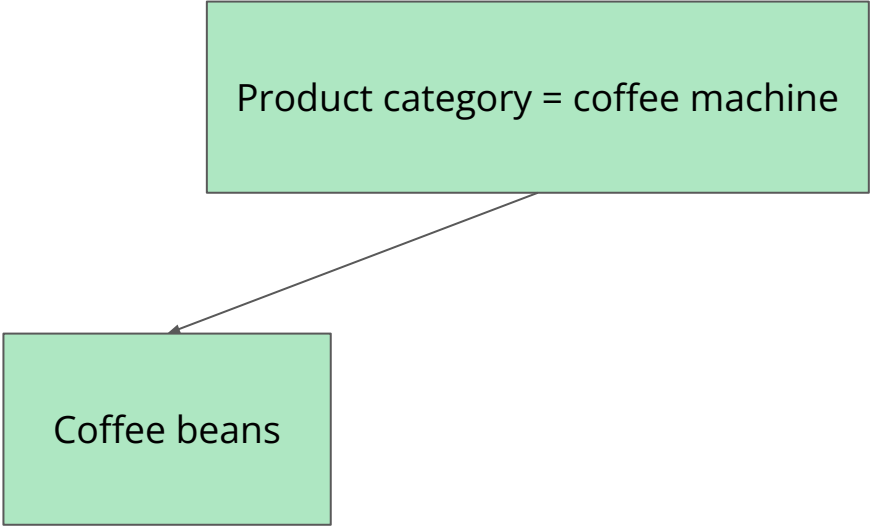
Product category = coffee machine

Training two towers model for complementary recommendations

During training we know the complementary categories from target products

But for inference we don't have targets, as we will be predicting them, so we don't know target categories

We can use predefined complementary categories mapping

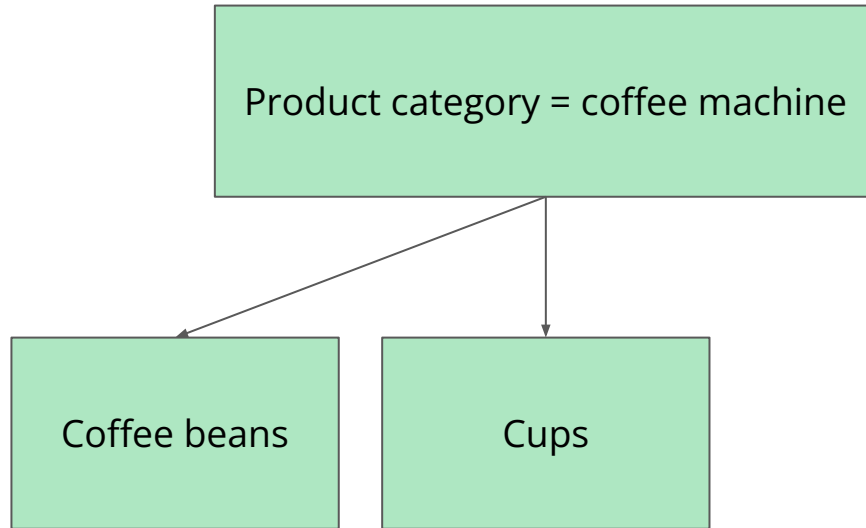


Training two towers model for complementary recommendations

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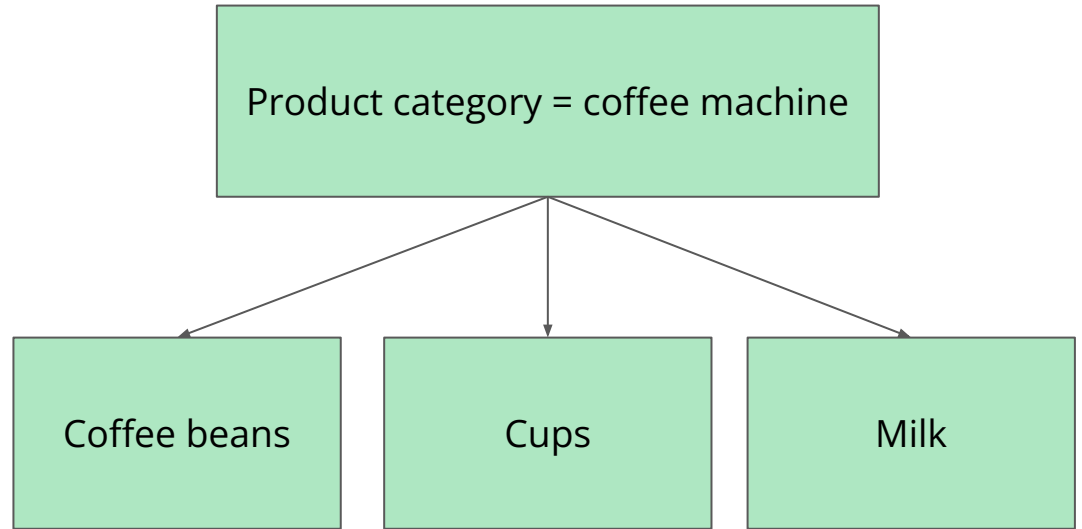


Training two towers model for complementary recommendations

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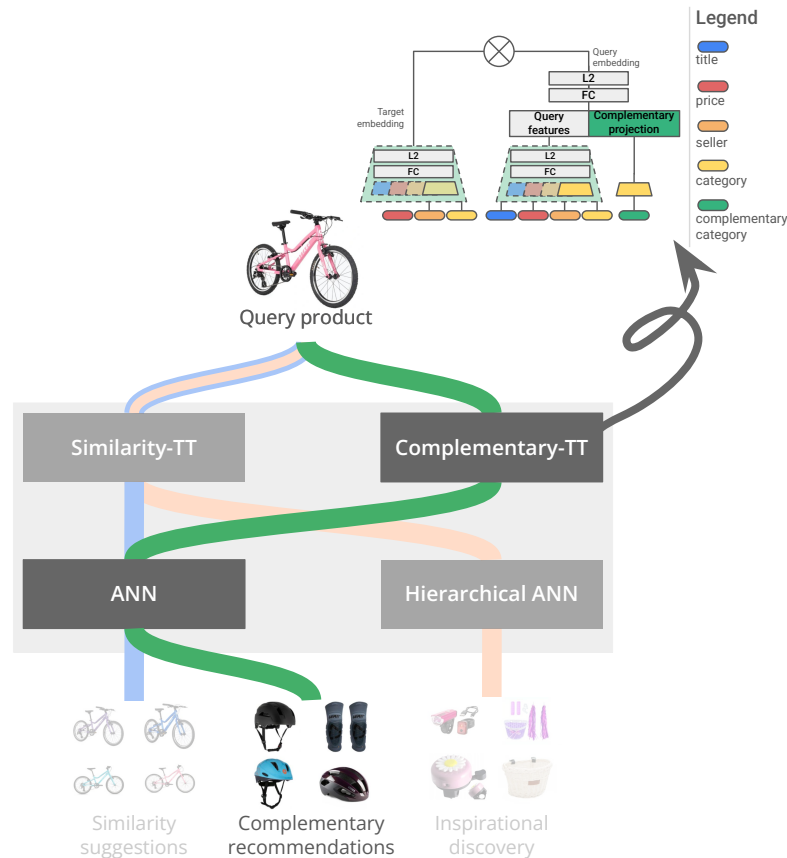
But for inference we don't have targets, as we will be predicting them, so we don't know target categories

We can use predefined complementary categories mapping



Complementary Recommendations - recap

- What are complementary recommendations?
- Training two towers model for complementary recommendations





What are inspirational
recommendations?

Inspirational Challenges

1. Balance novelty and relevance
2. Suitable for infinitive scroll
3. Minimize maintenance costs



Inspirational Challenges

Solutions

1. Balance novelty and relevance
2. Suitable for infinitive scroll
3. Minimize maintenance costs

1. **User topics** as an input and **interleaving** results

2. Sampling from **Hierarchical index**

3. Reuse **Similarity-TT**



Inspirational Recommendations

Example of real user browsing history:



210,00 zł

SMART

MEGA WIELKA
KOPARKA JEŹDZIK ...



5,99 zł

SMART

PALETKA MALARSKA
PALETA DO MIESZANI...



119,00 zł

SMART

KOPARKA GIGANT
LENA 50KG JEŹDZIK ...



4,24 zł

SMART

PALETKA MALARSKA
PALETA DO FARB MIE...



29,99 zł

SMART

MISKA
SPOWALNIAJĄCA JE...



10,99 zł

SMART

MISKA
SPOWALNIAJĄCA JE...

time

Inspirational Recommendations

User browsing history → Topics:



210,00 zł

SMART 

MEGA WIELKA
KOPARKA JEŹDZIK ...



5,99 zł

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Topic 1

Topic 2

Topic 3

Inspirational Recommendations

User browsing history → Topics → Diverse retrieval per topic:



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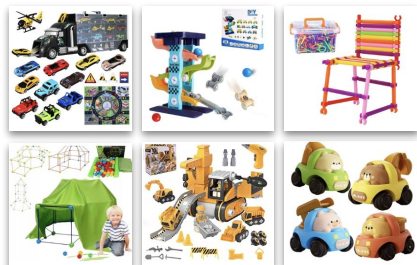


29,99 zł
SMART
MISKA
SPOWALNIAJĄCA JE...

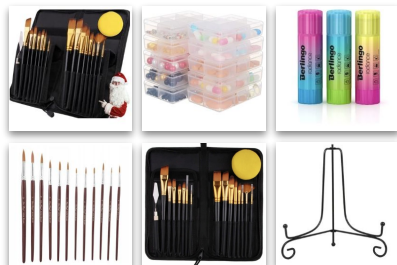


10,99 zł
SMART
MISKA
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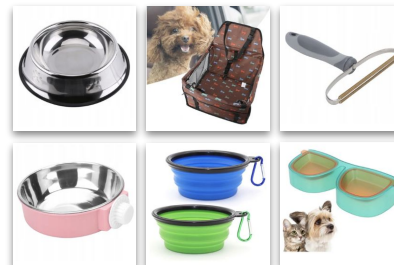
Topic 1



Topic 2



Topic 3



DIVERSE RETRIEVAL

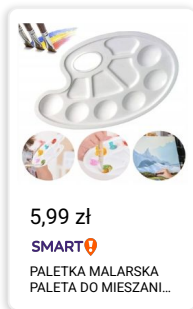


Inspirational Recommendations

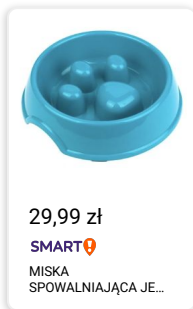
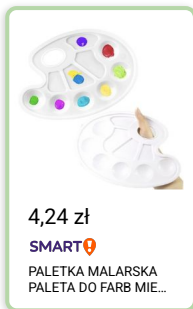
User browsing history → Topics → Diverse retrieval per topic → Recommendations:



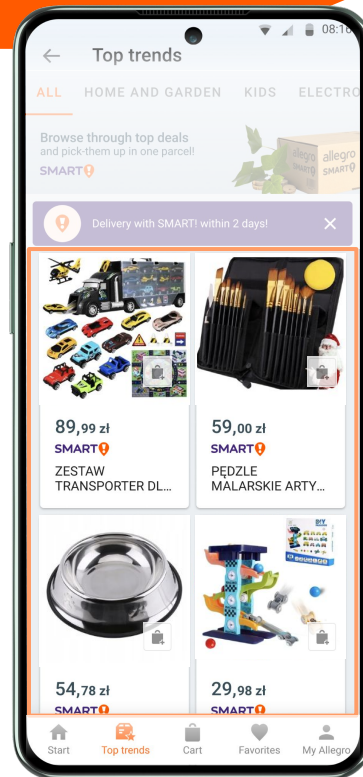
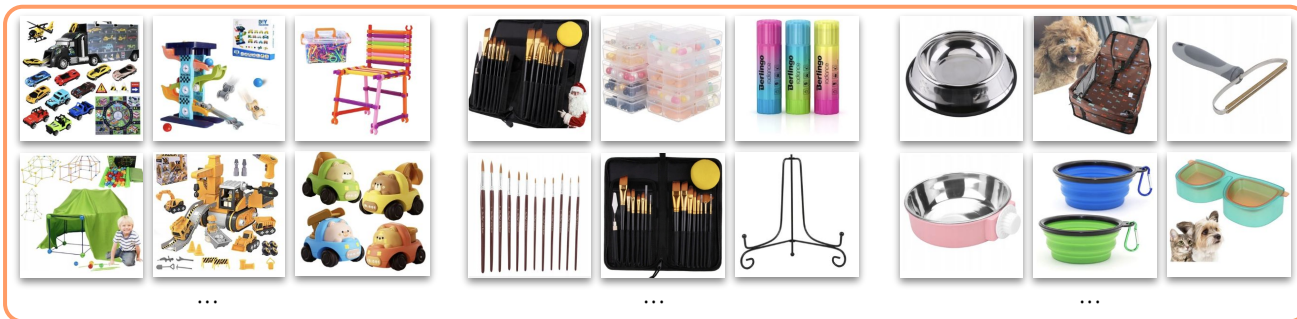
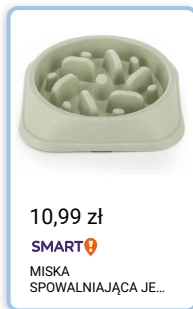
Topic 1



Topic 2

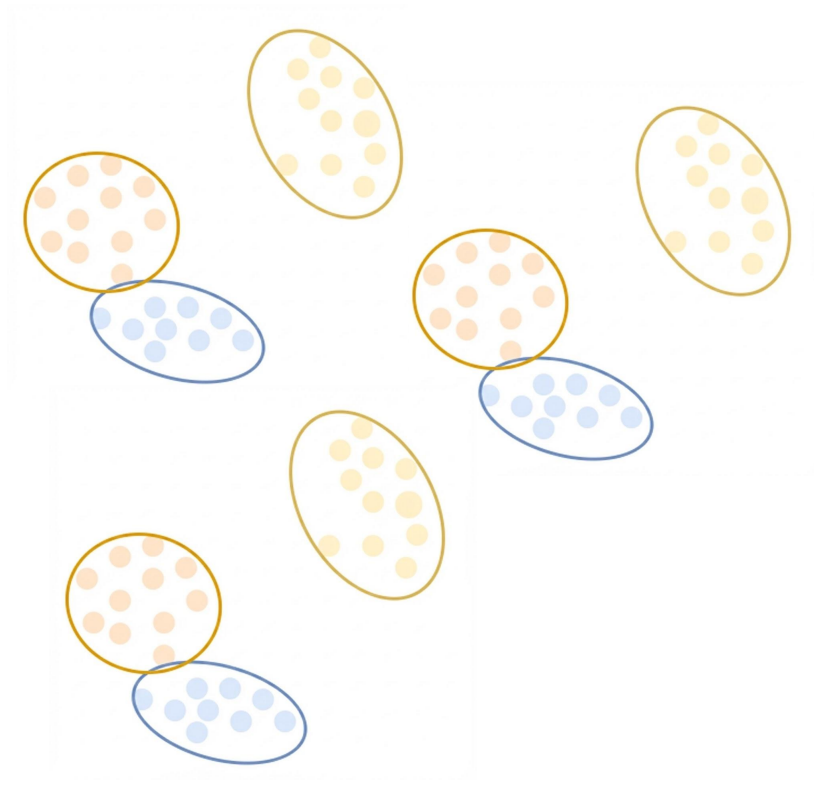


Topic 3



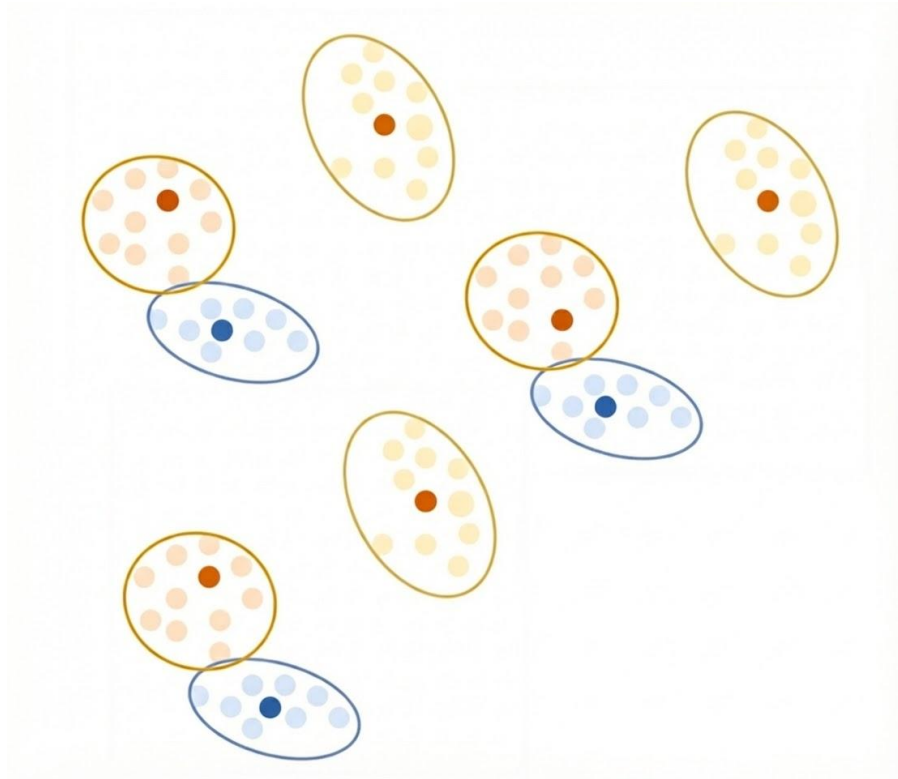
Diverse retrieval with hierarchical index

1. Cluster embeddings from Similarity-TT with K-means



Diverse retrieval with hierarchical index

1. Cluster embeddings from Similarity-TT with K-means
2. Insert each of cluster centroid into index



Diverse retrieval with hierarchical index

1. Cluster embeddings from Similarity-TT with K-means
2. Insert each of cluster centroid into index
3. During inference retrieve closets centroids then items in each cluster



Inspirational Recommendations

Encoder: Similarity-TT

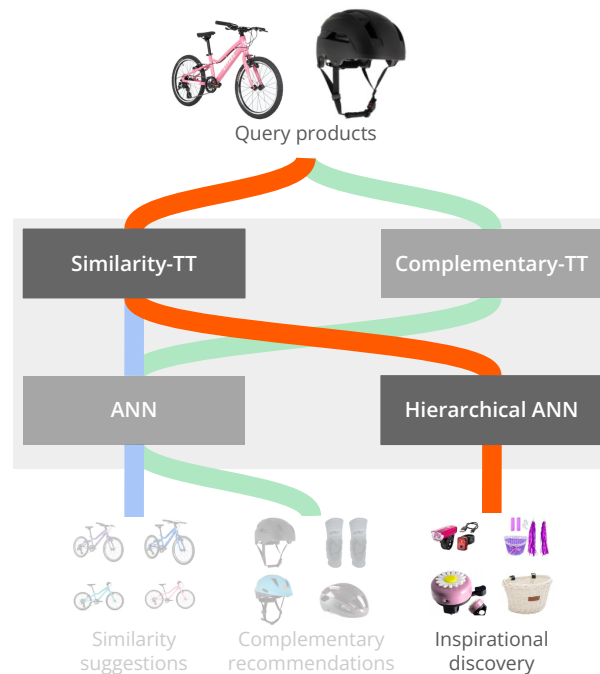
same as in similarity search

Indexing: ANN index

(+) query = recent clicks aggregated to category level (topics)

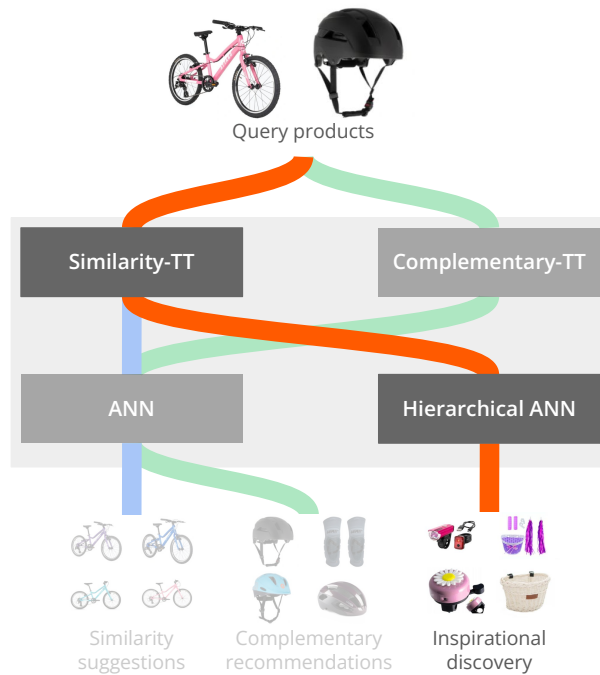
(+) index = K-means clustered embeddings from Similarity-TT

(+) interleaving between topics for increased diversity



Challenges

1. Hard to add new features and develop further
2. Need to cluster everyday and is limited by index sizes
3. Small control over content and diversity of results



Challenges

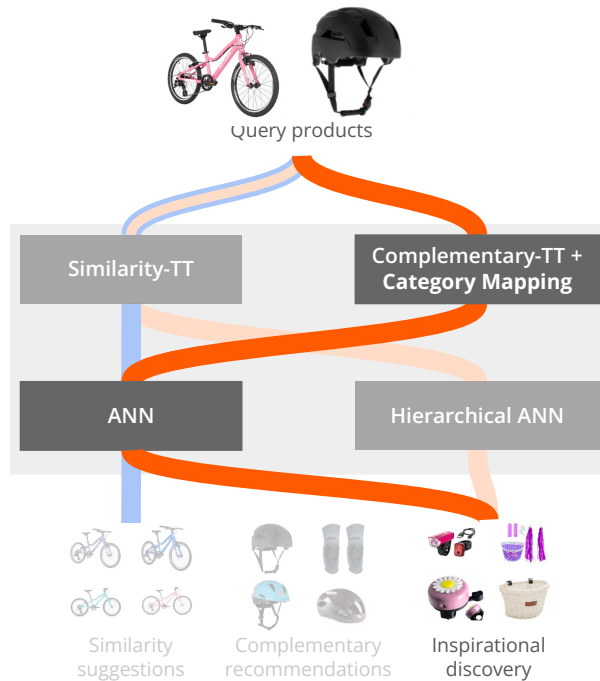
1. Hard to add new features and develop further
2. Need to cluster everyday and is limited by index sizes
3. Small control over content and diversity of results

Solutions

1. Use new **Category Mapping** as external knowledge

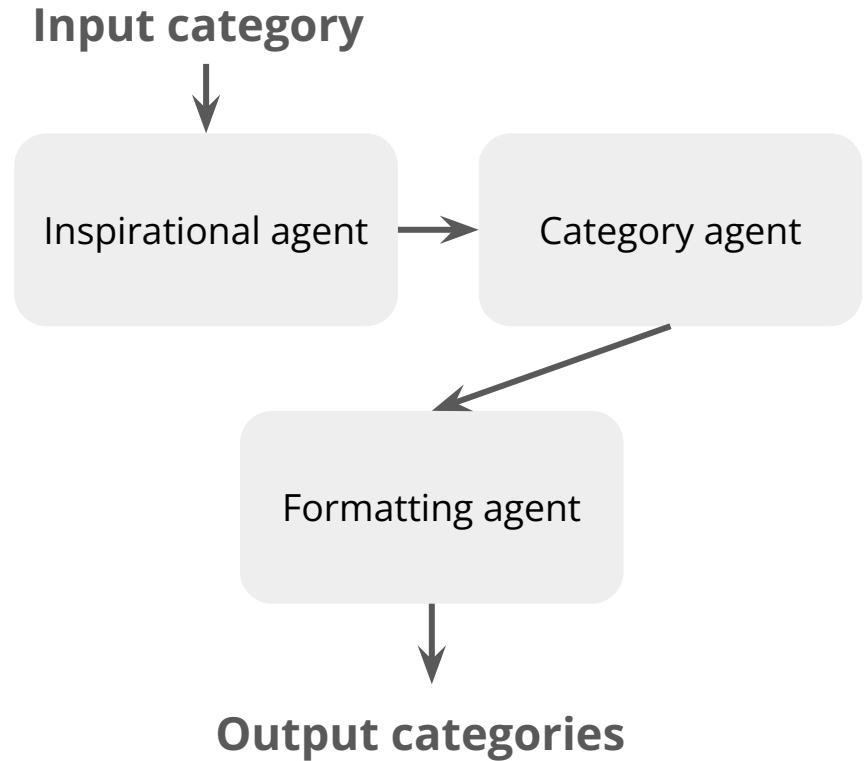
2. Mapping can be calculated **periodically**

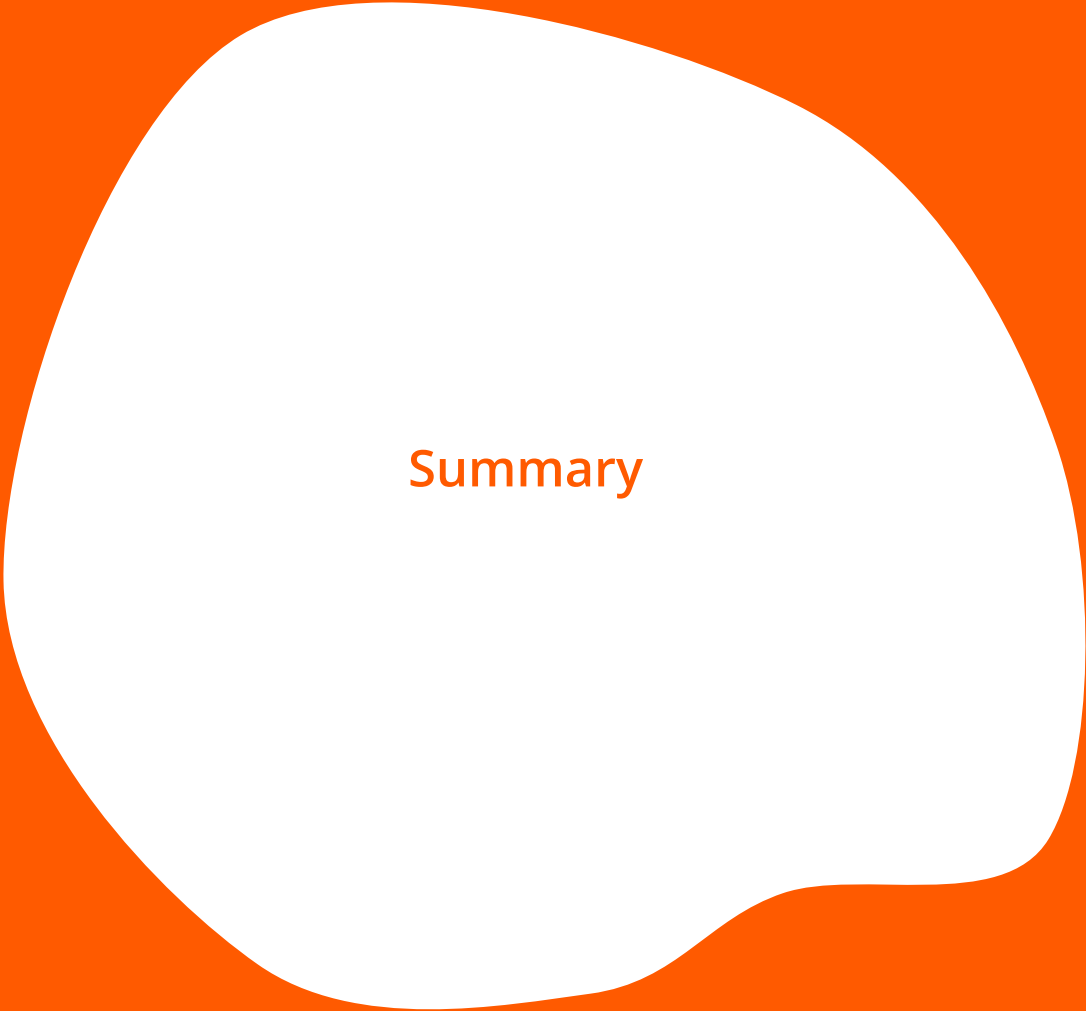
3. We define how mapping behaves



Category mapping with LLMs

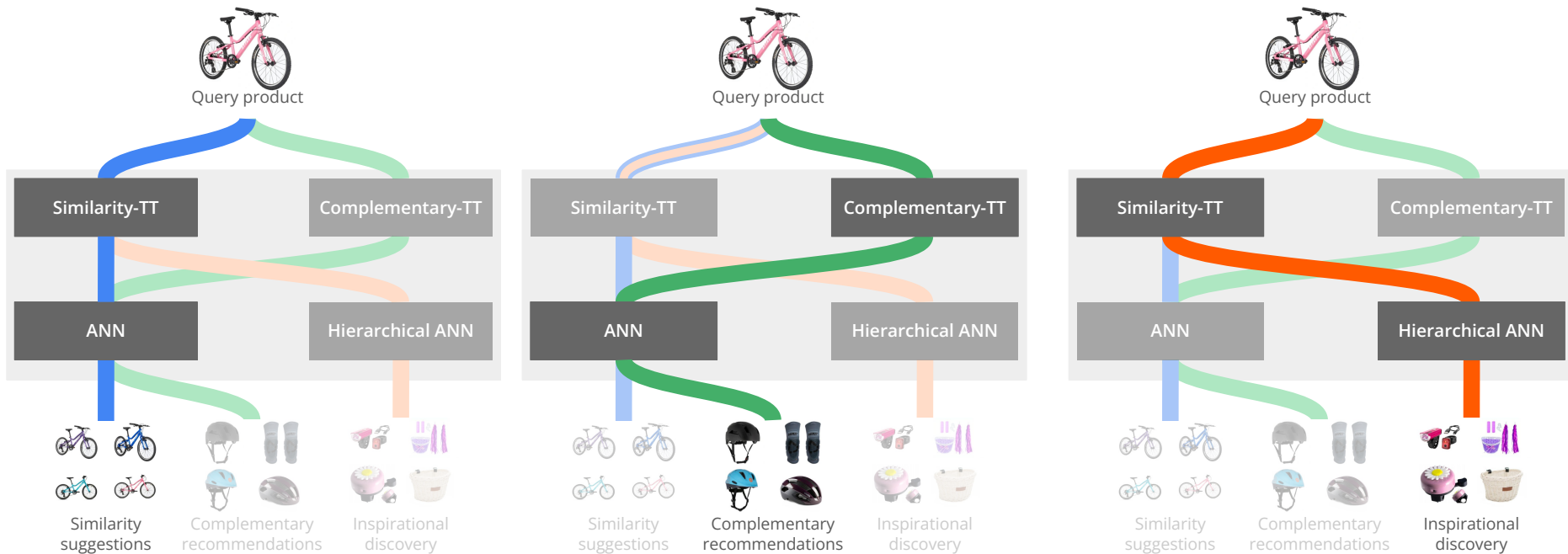
1. Inspirational agent
 - a. Given input category generates concepts of inspirational categories
2. Category Agent
 - a. Given concepts uses search tool to map it to allegro categories and assures quality
3. Formatting agent
 - a. Creates desire output format



A large, white, irregular blob-like shape is centered on a solid orange background. The blob has a soft, organic feel with rounded edges and a few small indentations. In the center of this white shape, the word "Summary" is written in a matching orange color.

Summary

One infrastructure to rule them all



Results

Solution is deployed at Allegro.com on wide range of placements across user path.

Scale:

- 160k sellers
- 20M active buyers
- 300M products

Throughput:

- 40ms latency (p99)
- >20k rps

Table 1: A/B test results of Similarity-TT and Complementary-TT on product page. Values indicate % change from the baseline, * denotes statistical significance at the 0.01 level.

model	mobile app		desktop	
	CTR ↑	GMV ↑	CTR ↑	GMV ↑
Similarity-TT	+2.11% *	+0.13%	+2.37% *	+0.29%
Complementary-TT	+1.62% *	+0.09%	+1.06% *	+0.31%

Table 2: A/B test results of introducing inspirational recommendations on product page (desktop traffic). Values indicate % change from the baseline, * denotes statistical significance at the 0.01 level, and the best result is bolded.

view	CTA ↑	CVR ↑	bounce rate ↓	exit rate ↓
carousel	+3.12% *	+1.38% *	-4.09% *	-1.82% *
infinite feed	+4.15% *	+2.22% *	-5.74% *	-1.66% *

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Next steps

1. Real-time **personalization**
2. **Foundational** models for content-based retrieval
3. Optimal **blending** strategies between organic and sponsored content

Questions?



Aleksandra Osowska-Kurczab
Machine Learning Manager



Tymon Huchla
Senior Data Engineer



Paweł Młyniec
Junior Research Engineer

our lab



paper



allegro
ML research